Battle Of The Blood Mimics

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Abstract

Photoacoustic (PA) imaging combines the optical absorption contrast of laser excitation with the acoustic resolution of ultrasound, enabling non-invasive visualization of blood flow and biomarkers.Standardised blood-mimicking fluids (BMFs) are needed to ensure reproducible testing and calibration of PA technology. This study compares two candidate BMFs, a 1% carbon-sphere fluid mixture and a 3.24% polystyrene-microsphere fluid mixture, to evaluate their performance in PA vector-flow imaging. Experiments were conducted across wavelengths from 680 to 880nm using a 2.95mm diameter vessel wall-less phantom. The 3.24% polystyrene suspension achieved a better signal-to-noise ratio (−6.7 ± 1.3dB), whereas the 1% carbon suspension delivered more accurate flow-velocity measurements (2.07 ± 0.30mm/s, within 7% of expected). These findings reveal a trade-off between signal quality and quantitative flow accuracy. While the carbon fluid offers better flow measurements, the polystyrene fluid provides superior imaging clarity. This study demonstrates why the development of standardised and reliable BMFs for photoacoustic systems is critical to future clinical translation and highlights the key trade-offs between potential BMF candidates.

1. Introduction

Photoacoustic (PA) imaging is an emerging imaging technique which pairs the high contrast of optical-absorption techniques with the penetration depth of ultrasound (US). PA imaging uses short near infrared light pulses (600-900nm) directed into tissue to produce acoustic waves from tissues of interest such as haemoglobin. These light pulses are absorbed by hemoglobin in red blood cells (RBCs) orders of magnitude more than the surrounding tissue. The absorption leads to a small but rapid thermoelastic expansion of the haemoglobin creating acoustic waves in the tissue. Due to acoustic waves experiencing 2-3 orders of magnitudes less scattering than light they enable PA imaging to happen at depths of up to 9cm. These acoustic signals are received by an ultrasound transducer at multiple angles which through data processing can measure flow speeds of blood down to 0.04mm/s. In addition, PA is spectroscopic in nature, by analysing their spectral signatures we can determine physiological parameters such as blood-oxygenation and glucose level. The high contrast, deep penetration, flow speed sensitivity, and spectroscopic nature of PA demonstrates its potential to revolutionise medical imaging for research and clinical applications.

PA positions itself as an information dense, non-invasive imaging technology that can be easily used by doctors to help diagnose and treat patients. PA can be used to identify adverse blood flow abnormalities such

as clots, issues within microvasculature, and cancer/tumor supporting angiogenesis. It can also be used for monitoring patient improvement with issues such as burn healing, tumor treatment, diabetic ulcer healing, and stroke recovery. With only 4 commercially available systems world wide it is clear that PA must be developed further before it is accepted into clinical practice.

To support the advancement of PA imaging, further testing and refinement of experimental materials is required. This study aims to determine which of two blood-mimicking fluids (BMFs), a 1% carbon sphere particle fluid or a 3.24% polystyrene sphere particle fluid, provides the most reliable performance in PA experiments. By comparing the signal to noise ratio (SNR), flow speed accuracy, and ease of use under a range of lasing wavelengths, this report aims to inform the selection of optimal testing materials and conditions. This research will contribute to the development of a robust testing protocol for future PA testing and development.

1.1 The Blood Mimicking fluids

Carbon particle fluids exhibit strong optical absorption due to their black pigmentation and glassy surface, producing a higher PA signal response and offering a cost-effective testing medium. However, these particles tend to fall out of suspension rapidly, and their reflective, glassy surfaces can also introduce image noise. Their variable particle size of 2-12μm compared to red blood cells (7.5 to 8.7μm) also limits physiological accuracy.

In contrast, polymer sphere fluids more closely mimic the optical scattering behavior of blood. The red pigment and non-glassy surface of the spheres reduce reflection-induced noise. This fluid also maintains uniform suspension during imaging. Despite providing more realistic optical behavior, polymer-based fluids typically yield weaker PA signals, can stain experimental phantoms, and are significantly more expensive than the carbon alternative.

2. Theory

To compare the performance of the blood-mimicking fluids (BMFs), it is essential to understand the principles of Doppler shift, velocity flow calculations, and signal-to-noise ratio (SNR). These parameters are used to evaluate how accurately each fluid replicates blood flow characteristics in photoacoustic imaging.

The Doppler shift describes the frequency change of an ultrasound signal reflected from moving particles within the medium. The maximum detectable velocity before aliasing occurs is given by:

$$V_{\max} = \frac{F_{nyq}\, C}{\cos(\alpha - \phi)\, F_0} \tag{1}$$

Where V_{max} is the maximum velocity through the blood vessel that won't be aliased due to the Nyquist theorem. F_{nyq} is the nyquist frequency, half of the frequency of the laser. C is the speed of sound through the medium. ϕ is the receiver angle and α is the vein angle from the normal to the ultrasound detector. Lastly, F_0 is the center frequency of the ultrasound detector. This equation calculates the V_{max} for our pump speed calculations based on known values. It also is the expected average velocity of the BMFs at the related pump speed.

The expected Doppler frequency shift for a given flow velocity can be expressed as:

$$F_d = \frac{V \times \cos(\alpha - \phi) \times F_0}{C} \tag{2}$$

Where F_d is the expected doppler shift for a given receiver angle. V is the velocity of the hemoglobin through the blood vessel. This equation provides a theoretical baseline against which measured shifts from the BMFs can be compared to assess accuracy. The size of this shift also depends on the velocity of the source so this equation can be rearranged to determine the flow speed of a fluid based on its frequency shift.

$$\text{Pump rate} = V_{\max} \times \frac{\pi d^2}{4} \times [\text{conversion constants}] \tag{3}$$

This is the equation for the pump speed used in testing. Where ‘d’ is the diameter of the vessel in the phantom and conversion constants are used to ensure the outcome is ml/minute.

Finally, the SNR quantifies the proportion of the doppler signal from moving particles to the background “clutter” produced by stationary particles, as shown in Figure X. Because clutter peaks are much larger than the doppler signal, SNR is expressed in decibels (dB). A less negative SNR value indicates a stronger doppler signal relative to the clutter present and therefore improved detection performance.

2.1 Predictions and Hypothesis

TABLE 1: Known parameters

Parameter	Value
Pulse Repetition Frequency (PRF)	20 Hz

Nyquist Frequency (f_{nyq})	10 Hz
Centroid Frequency of the Transducer(f_0)	7.6 MHz
Receiving Angles (ϕ)	-10°, -5°, 0°, 5°, 10°
Speed of Sound through phantom (C)	1449.3 ± 3.37 m/s
Vein angle from the normal to the ultrasound detector (α)	90°

To determine which BMF is best the following predictions were made to compare to the results:

Predictions were made for the max velocity using all receive angles, below is an example for a 10° receive angle. V_{max} is smallest at the largest receive angle, thus ±10° results in our limit for aliasing to occur.

$$V_{\max} = \frac{F_{nyq}\, C}{\cos(\alpha - \phi)\, F_0} = \frac{10 \text{ Hz} \times 1449.3 \text{ m s}^{-1}}{\cos(90° \pm 10°) \times (7.6 \times 10^6) \text{ Hz}} = 10.98 \text{ mm s}^{-1}$$

The pump speed should be set to deliver at most half the max velocity of flow to ensure no aliasing occurs in the results due to the Nyquist theorem. In addition, this study's primary focus is slow blood flow since this will be tested extensively by PA researchers. The pump speed was set to deliver a velocity closer to that experienced in a small vessel, putting the expected average velocity far from aliasing. 1.936 mm/s was chosen as a value that represented the slow blood flow in tissue that would likely be of interest in future clinical studies.

$$\text{Pump rate} = 1.936 \text{ mm s}^{-1} \times \frac{\pi \times 2.95^2 \text{ mm}}{4} \times \left[60 \frac{\text{s}}{\text{min}} \times (1 \times 10^{-3}) \frac{\text{cm}^3}{\text{mm}^3}\right]$$

$$\text{Pump rate} = 794.1 \pm 3.2\ \mu\text{L/min}$$

The expected doppler shift was calculated for each receive angle, below is an example at ± 10° receive angle.

$$F_d = \frac{V \times \cos(\alpha - \phi) \times F_0}{C} = \frac{1.936 \text{ mm s}^{-1} \times \cos(90° \pm 10°) \times (7.6 \times 10^6) \text{ Hz}}{1449.3 \text{ m s}^{-1}} = \pm 1.763 \text{ Hz}$$

Lastly, this study predicts that the 1% carbon fluid will have a larger SNR due to its optical properties producing a larger signal.

3. Literature Review

The earliest documentation of the PA effect being used in technology was 1880 by Alexander Bell. After the invention of the telephone Bell decided to turn his attention to light transmitted information. By June 1880 Bell and his assistant Charles Sumner Tainter were able to transmit voice messages roughly 213 meters using focused sun rays and a vibrating mirror as their 'laser'. This was done using the photosensitivity of selenium as a receiver which was connected to a powered telephone circuit. Through the research on his "Photophone" Bell made the key discovery that; by illuminating solids with a pulsed beam of light, the energy created an acoustic wave at the same frequency as the pulses. This discovery astonished the scientific community of the time but as communication through radio developed through the early 20th century the idea of a photophone was all but forgotten.

Revived in the late 1930s for infrared gas analysis, the photophonic effect saw brief renewed interest before being replaced by more sensitive techniques like gas chromatography and infrared spectrophotometry in the late 1940s. Once again, after falling to the wayside for decades, the photophonic effect had a resurgence in the 1960-70s due to the discovery of laser technology in the 1960s. Lasers provided a power and stable light source which paved the way for PA spectroscopy in the coming decades. The earliest case of biomedical applications for photoacoustics was in 1964 by French scientist Amar et al. They discovered ultrasound signals were generated in the occipital bone of a rabbit when low power light (non-injuring) was pulsed through the rabbits eye. At the time Bell labs was focused on the interaction of light with solids. This is when Rosencwaig coined the term 'Photoacoustics' to be used referring to the phenomenon interacting with solids in his 1973 paper. Early biological discoveries using PA followed, such as, confirming early hypotheses about the proteins used in cataract development. As PA imaging advanced two modalities emerged: photoacoustic computed tomography (PACT) and photoacoustic microscopy (PAM). PACT and PAM provide a trade-off between depth and resolution. PACT uses diffuse illumination and parallel detection with a transducer array, which is ideal for deep, real-time preclinical imaging, such as tumors, despite its low resolution (200–500 µm). In contrast, PAM offers high-resolution, microscopic detail. It uses a confocally aligned optical-acoustic setup that mechanically scans point-by-point, deriving lateral resolution from the joint foci and depth information from acoustic time-of-flight. Both of these modalities are used for tracking blood flow, this study uses PAM as it focuses on blood flow and BMFs which need microscopic detail. The first records of using the doppler shift technique to track blood flow that this study could find was by Lihong V. Wang and his team in 2008. In comparison blood flow has been tracked by ultrasound since the early

1960s, this shows how young PA imaging of blood flow is. Throughout history researchers have used non-stardised BMFs, reducing the reproducibility, clarity and capability for collaborative work thus slowing PA technology development.

Research [] explained in 2023 that for PA to transition from a 'nice-to-have' to a 'need-to-have' modality it will require clinical trials focusing on the value for patients and clinicians. This is why clinical studies are currently focusing on the use of PA to address gaps in identification of cancer. For PA to solidify itself as an imaging modality it must address this niche better than existing options, likely through being; cost effective, portable, increase the success rate of identification, and be easy to operate in many healthcare settings. Current leaders in PA imaging, such as Luxonus from Japan, are focused on using the increased resolution and ease for patients to create a commercially viable product able to be installed in offices to be used for advanced diagnosis and monitoring. The LME-01 from Luxonus uses a unique semi sphere of ultrasound sensors allowing for signals to be received at greater angles thus allowing for vessels traveling at 0°- 60° to parallel to be imaged, creating highly detailed images. Seno medical's Imagio is a PA breast cancer diagnostic tool that has been FDA approved. This approval allows PA technology to be in the hands of doctors and help expose patients to PA technology, this will help the transition in clinical settings for future PA companies. The Imagio imaging device focuses on being an effective clinical tool by using a familiar shaped ultrasound wand and simply overlaying a vessel map over an ultrasound image to help identify tumors in patients' breasts. Vevo F2 LAZR-X20 is a preclinical research tool focusing on versatility, by tuning the laser it can have a max resolution of 30 micron or an image depth of up to 9cm. All of these leaders are focused on blood flow, and blood oxygen vessel mapping to focus on cancerous tumor identification and monitoring.

Development is ongoing for LED powered PA imaging technology which delivers a safe and portable monitoring device, recording things such as; blood biomarkers, arthritis, and cancerous growths. This is promising for PA as it means in some instances LEDs can be used which are much cheaper, and reserve the expensive high powered lasers for more specialised diagnosing situations. It has been shown that CNM-based nanocomposites can be used as injectable contrast agents to broaden the utility of PA imaging by absorbing light across the UV, visible, and NIR-I/II windows. This in turn enhances image resolution, increases penetration depth, and enables more versatile multispectral analysis allowing for the validation of spectroscopic techniques. This emphasized photoacoustic intensity has proven effective for in vivo monitoring applications like drug delivery and tumor cell targeting. This shows that for cases which need more intensity, PA can adapt to be a semi-invasive modality to fulfil those needs. Lastly, future advancements with machine learning and AI appear promising for increasing diagnostic accuracy but need further development.

BMFs are very important as they offer a standardised and repeatable test that can be used to calibrate and compare machines. Calibration is required for any tool entering clinical use as inaccurate/uncalibrated tools can threaten the safety of patients as they can lead to misdiagnosis. Table 2 is used when making ultrasound mimics so it is missing the crucial component for PA, optical absorption. It has also been shown that the Table 2 values do not exactly match with human blood.

TABLE 2: Properties and the values that BMFs should aim for as defined by the International Electrotechnical Commission.

Properties of BMFs	Values
Viscosity ($\times 10^{-3}$ Pa·s)	4.0 ± 0.4
Attenuation (dB/cm/MHz)	≤ 0.1
Acoustic speed (m/s)	1570 ± 30
Density ($\times 10^{3}$ kg/m^{3})	1.050 ± 0.040

There have been a variety of BMFs used in history; in 1994 india ink was used in an early phantom to demonstrate the possibilities of PA in medical imaging. Since then limitations have been found with india ink, such as brighter signals at the edges of vessels, and providing flat absorption spectra. Nano particles in suspension are now used widely in PA research to emulate RBCs in blood. There is limited research on direct comparisons between BMFs and human blood. This leaves it unclear if the BMFs are improving in terms of accuracy to theoretical predictions or accuracy to how human blood would behave. Both carbon and polystyrene micron sized spheres have been used in suspension to mimic blood. These fluids must attempt to match the properties mentioned in table 2 in addition to having large SNR, accurate flow speed measurements and stability in suspension. Another unconsidered aspect when using micron spheres is the shape, RBCs are a biconcave disc (7.5 to 8.7µm diameter and 1.7 to 2.2µm in width) which means light would be interacting with it differently based on its orientation, unlike a sphere. Even though there are three main cells in human blood, platelets (~2µm), white blood cells (WBC) (12-15µm) and RBC . PA light waves are tuned to be absorbed mostly by hemoglobin which is only present in RBC, thus the artificial particles should focus on replicating RBCs and not other cells. Future advances could include manufacturing red

polystyrene biconcave discs to better mimic the optical and acoustic properties of RBCs.

Despite progress in photoacoustic imaging technology, several critical gaps remain in blood mimicking fluid development that limit reproducibility and standardization of PA imaging research.

A major challenge is BMF stability during experiments. When the density of particles is more than that of the surrounding liquid, settling may happen, which can be visible at low flow speeds. An example of a particle that behaves this way is Sephadex. Polystyrene scatter particles can precipitate to the bottom when their density is greater than the mixture fluid. This density driven settling creates concentration gradients that require constant remixing and introduce measurement inconsistencies.

There is currently no universal standard for BMF composition across laboratories. Current standardised methods for system testing are lacking, impeding objective device performance evaluation, calibration and inter-device comparisons. Different groups use vastly different formulations, from India ink solutions to polystyrene microspheres to custom nanoparticle suspensions, making cross-study comparisons difficult. IPASC has developed recommendations for ideal phantom characteristics to help standardize practices and improve system calibration across the field. Without agreed-upon reference fluids with well characterized properties, it's difficult to establish benchmark performance metrics or determine whether reported differences in system performance reflect actual technological advances or simply variations in test fluids.

Real blood shows wavelength-dependent absorption that changes with oxygenation state, which is fundamental to many PA applications. To test this, it has been found that using different concentrations of intralipid (a fat emulsion) can mimic both deoxygenated and oxygenated blood. Several general limitations have been identified in prior phantom work - tissue phantom optical and acoustic properties were often not characterized or were not biologically relevant, and test methods using ex-vivo tissues as phantoms lack reproducibility. Linear spectroscopic inversions are widely used in photoacoustic imaging to estimate blood oxygen saturation, but they don't account for the spatially varying wavelength-dependence of the light fluence within tissue, introducing spectral coloring as a potentially significant source of error.

Using animal blood to mimic human blood comes with a number of issues. Using animal products including blood must first be approved by ethics committees creating a barrier for use. There is extensive research about safety precautions when using animal blood as mishanding it can result in disease, hospitalization, and even death. Animal and human blood have a shelf life for the hemoglobin proteins it contains and the viscosity which would affect flow speed, this varies drastically depending on storage conditions and what animal it is taken from. Future research into artificial blood products could pose another option for tests as they have extended shelf life, up to 2 years at room temperature. Lastly, animal blood has different sized RBC and hemoglobin concentrations to humans which is pivotal for the signal strength in PA imaging.

There is no agreed upon standard for BMF testing. Comparing flow speeds and doppler shifts to expected values is an effective analytical way of comparing fluids, but comparisons to human blood results would be ideal. It has been shown that using the same poly-particle fluid as this study with a range of vessel widths the flow results consistently underestimate the known velocities. Tests for blood flow using BMFs use a variety of experimental setups, from tubes suspended in water to more ideal wallless phantoms. Each method can lead to varying results highlighting the importance of standardisation of both phantoms and fluids.

BMFs are essential for PA development as they allow for consistent, repeatable, and collaborative experiments. The lack of research into BMFs has left the PA research community lost. Without standardisation of accurate and stable BMFs and phantoms researchers are unable to relate their data to those around them, limiting the pace of progression towards clinically viable PA technology. Future progress will depend on the creation of reference BMFs that replicate the optical absorption and acoustic properties of real blood, enabling reliable calibration and clinical translation of PA systems.

4. Experimental Methods

3.1 Phantom Fabrication

The initial step involved fabricating phantoms for vector flow testing. Sections of tin-foil were cut and shaped into rectangular moulds for the phantoms by folding up the corners and securing them with staples. Small holes were punctured on opposite sides of each rectangle (width-wise) using a copper wire. Connectors were inserted into these holes by lightly rotating them. A copper wire was then threaded through the length of the moulds by passing it through the connectors at each end. This tin-foil assembly-consisting of a rectangular bowl with connectors at both ends and a wire running longitudinally through the center-served as the final mold for the phantoms. Three moulds were prepared: two containing 1 ± 0.005 mm diameter wires and one containing a 2.95 ± 0.005 mm diameter wire. The latter was constructed with one connector due to the wire being too large to pass all the way through it, and protruded without a connector at the opposite end as shown in figure X. I think it would be a good idea to draw a diagram for this rather than adding a photo, dimensions of our phantoms/the mold would be good too.

An oven was preheated to 200°C. During the heating period, the phantom material (Humimic SimuGel™ #0) was cut into small cubes and placed in glass beakers for melting. Once the oven reached 132℃ , the beakers containing the cubes were placed inside and left for approximately 30 minutes to fully melt. The molten gel was then poured into the prepared molds.These were immediately returned to the oven for one hour or until

all bubbles had escaped the gel or were at the surface. Once at the surface they can be popped with a fine syringe needle.

Once the bubbles were removed, the moulds were placed on the bench to cool overnight. Once cooled, the tin-foil was carefully peeled away from the phantoms, ensuring no residual foil remained. The metal wire was then slowly extracted from the center of each phantom using pliers to grip the rod. The resulting product was a rectangular phantom with connectors and a hollow "vessel" running through the middle, as shown in Fig. X. This channel would serve as the conduit for fluid flow during testing.

3.2 Experimental Setup

All photoacoustic vector flow measurements were conducted using the 2.95 ± 0.005mm vessel phantom described in Section 3.1. The experimental configuration is illustrated in Figure X.

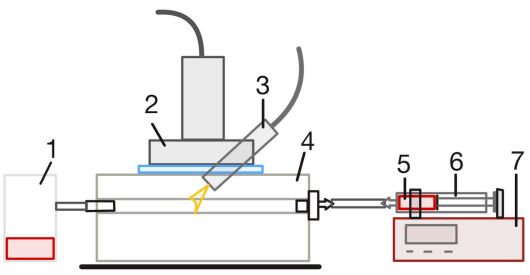

Fig.x - Diagram of Experimental Setup

3.2.1 System Components

The experimental apparatus consisted of seven primary components (Fig. x):

1. Fluid reservoir - A beaker positioned to collect outgoing fluid from the phantom.
2. Ultrasound transducer - Used for ultrasound signal detection and imaging.
3. Laser system - Opotek RADIANT 532 LD Tunable Optical Parametric Oscillator Laser, provided lasing at wavelengths ranging from 680nm to 880nm.
4. Flow Phantom - As described in section 3.1, containing the channel mimicking a blood vessel.
5. Blood-mimicking fluid - One of two blood mimicking fluids used, 1% carbon sphere particle fluid or 3.24% poly sphere particle fluid.
6. Fluid delivery syringe - Connected to the phantom inlet.
7. Syringe pump - Programmable pump for precise flow rate control.

3.2.2 Flow Rate Calibration

The syringe pump was set to deliver fluid at a rate of $794.1 \pm 3.2 \mu L/min$. This flow rate was calculated using the equations stated in the theory section. The internal diameter of the syringe vessel (12.25 ± 0.005 mm) was required to calibrate the pump.

3.2.3 Laser Alignment Procedure

Prior to alignment of the laser the power produced by the laser at each wavelength was measured.

The laser was aligned to maximise the photoacoustic signal from the vessel within the phantom. With the system running and the laser lasing at 680nm, the laser position was adjusted repeatedly while monitoring the peak signal amplitude. This was done until a maximum intensity was identified, indicating optimal overlap between the laser beam and the vessel. Once in position, the laser was secured to maintain alignment throughout the experiment.

The optimal alignment yielded peak signal amplitudes of approximately 220 bits for the microsphere-based fluid and 8000 bits for the carbon-based fluid, reflecting the difference in optical absorption between the two fluids.

3.2.4 Data Acquisition

For each blood-mimicking fluid, the following system was used:

1. The syringe was filled with the test fluid and placed in the syringe pump.
2. The pump was activated to establish steady flow through the phantom.
3. Once stable flow was confirmed, photoacoustic data was acquired at six wavelengths, 680nm, 720nm, 760nm, 800nm, 840nm and 880nm.
4. At each wavelength, 1000 frames were recorded using the provided acquisition script.
5. After completing all recordings, and before the next fluid was used, distilled water was pumped through the vessel to flush the phantom of any residual BMF.

All components remained fixed in position throughout the experiment to ensure consistent measurements and geometry across all trials.

5. Data Processing

In this section we will outline the procedure undertaken to acquire all key results found within our experimental data. Throughout this experiment MATLAB was used for processing all data obtained for all wavelengths measured for each fluid to find all key values from this experiment. Google Colab was also used to analyse frequency spectrum graphs to obtain values for SNR and doppler shift.

4.1 Image Reconstruction, Frequency Spectrum Graphs and Laser Intensity

First each data set was compiled using the code file Reconstruct_JS_RunPA.m. Within this file the images were set to the transducer's parameters and the image grid was set to produce an image at depths 15 to 20mm and lateral positions of -4mm to 4mm relative to the transducer center. This produced a 1000 frame image reconstruction of the fluid flow within the phantom vessel as the chosen photoacoustic fluid was being pumped through:

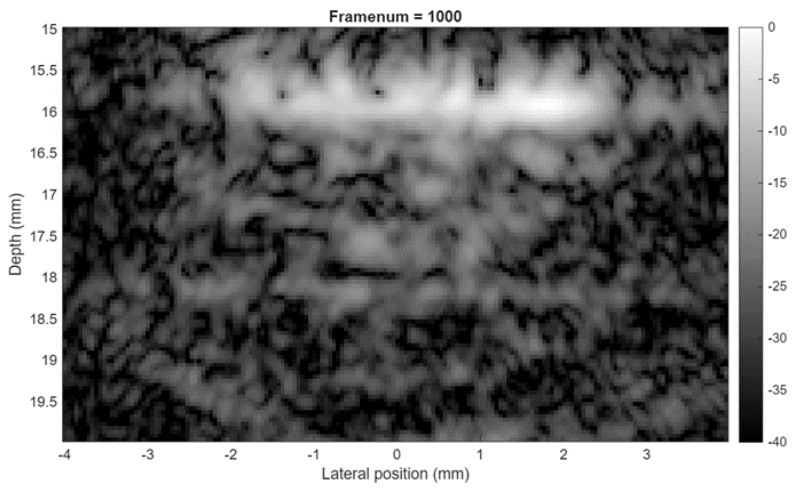

Figure A: Example image reconstruction of signals received via the transducer over all the receiving angles at each wavelength from 680-880nm

The brighter areas represent regions of higher reflectivity from the laser signal. The depths of 15-15.75mm and 18.5 to 20mm represent the noise produced by the phantom structure and the region at depths of 16-18mm represent the inside of the vein where the photoacoustic fluid was present. The bright lines found between 15.5-16mm and 18-18.5mm depths, represent the vein walls. These are present due to the large change in impedance moving from the humimic phantom gel to the chosen photoacoustic fluid. Lastly this produced frames_param_(fluid name and wavelength).mat files that were used in all other code files to obtain results.

Next we wanted to find the maximum intensity present in our reconstructed image to determine laser intensity. By using an altered version of the previous code to give the maximum intensity over the selected region, the file used was called Intensities_ Reconstruct_JS_RunPA.m. The region chosen was set at depths between 15.5 - 17.5mm and lateral positions set between -1.5 - 3mm, this can be seen as the bright region in Figure A which marks the top of the phantom vessel. The maximum intensity was found over this entire region and compared with the laser power found during the experimental procedure.

4.2 Doppler Shift Analysis

The frames_param_(fluid name and wavelength).mat files were then processed using the code file slow_time_spectrum.m to first isolate a desired region to obtain frequency spectrum graphs for each receiving angle. The chosen region was set as a 40 by 20-pixel box set with its top left corner set at pixel position (70, 36).

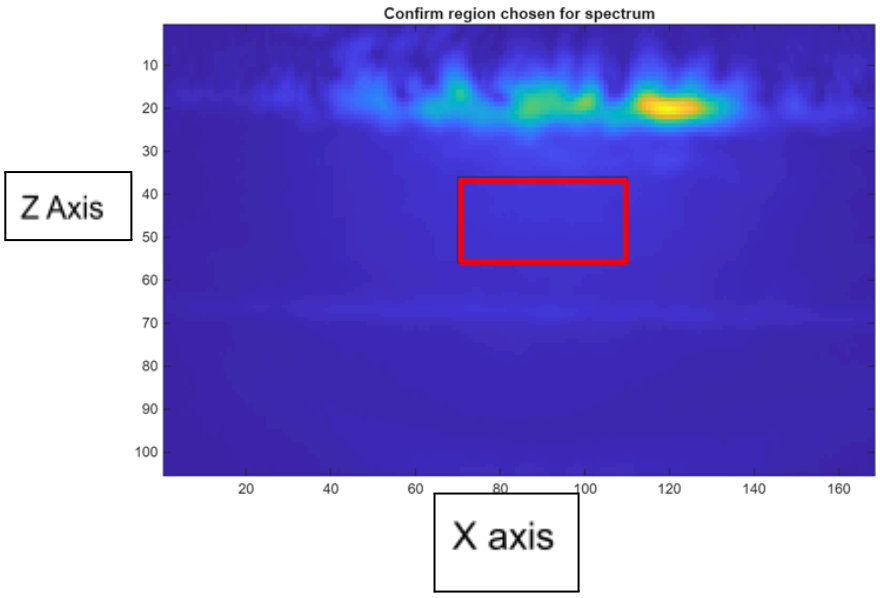

Figure B: Example image of chosen region, shown by the red box,for to plot frequency spectrums of the amplitude at each receiving angle for each wavelength from 680-880nm

The brighter regions show, as also seen in Figure A, the level of laser signal reflectivity due to impedance changes.

This code file then produced five frequency spectrum graphs each corresponding to a given receiving angle for this chosen region:

10°

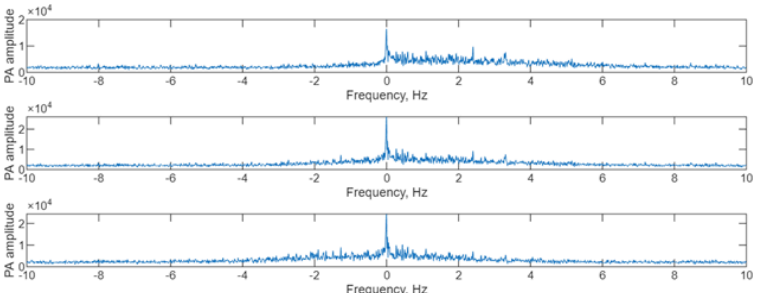

Figure C: Example frequency spectrum of the signal and noise amplitude at each receiving angle from 10° at the top to -10° at the bottom. The noise signal known as the clutter peak is found around a frequency of 0Hz while the signal is found across the spectrum giving much lower amplitudes by comparison. Frequency ranges from -10Hz to 10Hz

The Peak at approximately 0Hz is the clutter peak which represents the noise produced by stationary particles within our chosen region. The surrounding signal envelope represents the signal produced by the photoacoustic fluid flowing through the phantom.

In order to analyse the signal envelope further we needed to filter out the clutter peak. To do this we used a high-pass filter by integrating the code file highpass_GR_US.m to the slow_time_spectrum.m code file. The filter was set to act over the same region that the original spectrum was obtained from, with a cut-off frequency of 0.01Hz giving the following filtered frequency spectrums:

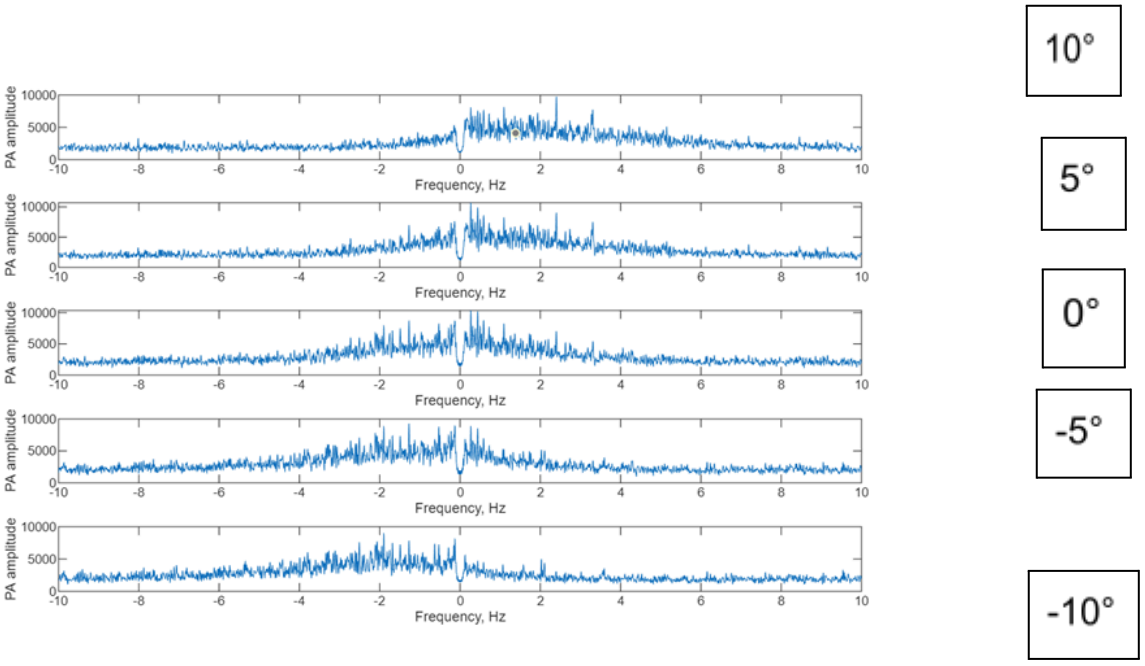

Figure D: Example 0.01Hz high-pass filtered frequency spectrum showing only the remaining signal envelope used to find signal amplitude and doppler shift. Frequency ranges from -10Hz to 10Hz.

The data contained within each graph was then exported into Excel using the code file figure_to_excel_code.m, these values could then be used in Google Colab to analyse for the Doppler shift at each angle.

By reconstructing the filtered frequency spectrum using plt.ployfit we can draw a precise estimate of the signal envelope of each data set. We then found the peak of each curve to find the frequency associated which represented the Doppler shift at the chosen receiving angle:

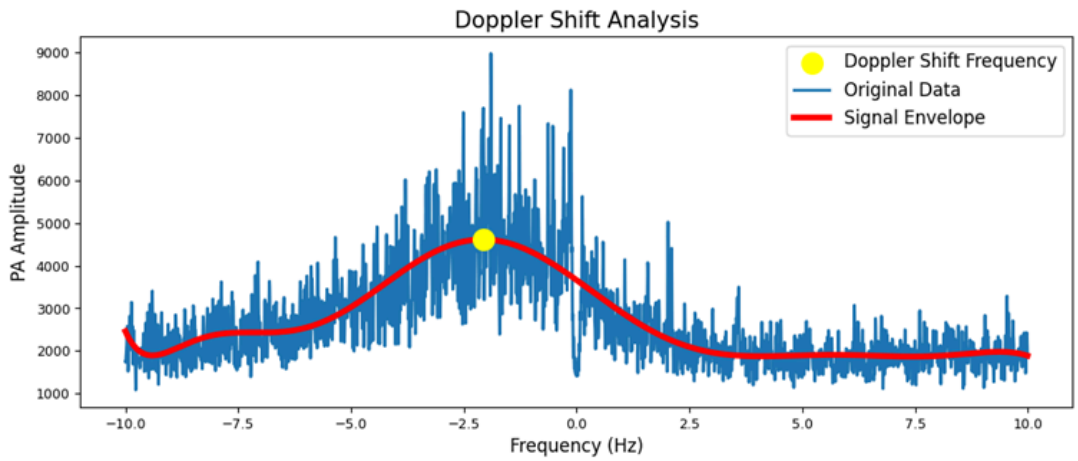

Figure E: Example reconstruction of 0.01Hz high-pass filtered frequency spectrum with envelope. The blue curve represents the original filtered data, the red line represents the fit for the signal envelope and the yellow dot represents the doppler shift frequency which is found by finding the peak signal amplitude.

Doppler shift frequencies were taken at every receiving angle for each data set, next the average doppler shift frequency at each angle was found for both fluids. These frequencies were then compared for each fluid to the expected doppler shift at the matching receive angle. This was found using equation 2.

4.3 Signal to Noise Ratio (SNR) Calculations

By also exporting the unfiltered doppler shift graph data to excel, shown in Figure C, we were able to reconstruct this data in Google Colab to find both the clutter peak amplitude and the signal envelope amplitude for each data set:

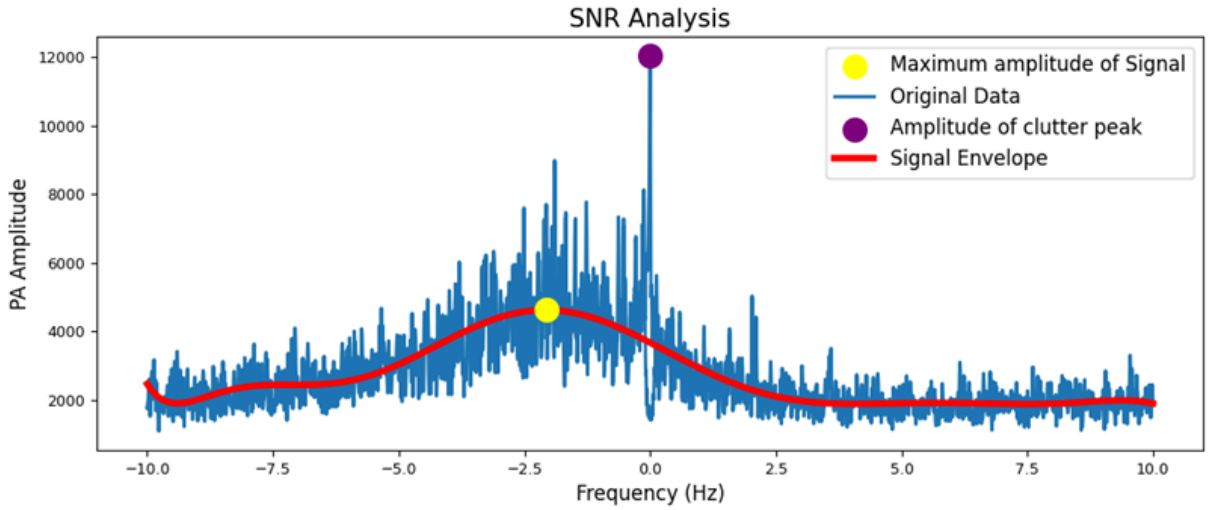

Figure F: Example reconstruction of unfiltered frequency spectrum with signal and clutter peaks marked by the yellow and purple dots respectively

To find the SNR of each frequency spectrum the amplitude values obtained for clutter peak and signal envelope peak were used in the following formula:

$$(SNR)_{dB} = 10\,log\left(\frac{A_S}{A_C}\right) \tag{4}$$

Where $(SNR)_{dB}$ is the signal-to-noise ratio scaled in decibels, A_S is the peak signal envelope amplitude and A_C is the amplitude of the clutter peak. Each SNR value was averaged over all angles for each laser wavelength measurement, completed for both photoacoustic fluids, and the standard deviation was found.

The SNRs found at each wavelength were then compared, to see which fluid gave a less negative value meaning the signal envelope amplitude was closer to the clutter peak amplitude therefore an image that contained less noise.

4.4 Vector Flow Analysis

By processing the frames_param_(fluid name and wavelength).mat file using the code file Compute_Vectorflow.m we were able to find mean and max velocities for each data set along with an associated standard deviation. First the code analysed all the given data in the frames_param_(fluid name and wavelength).mat file to obtain vector flow map for the reconstructed image:

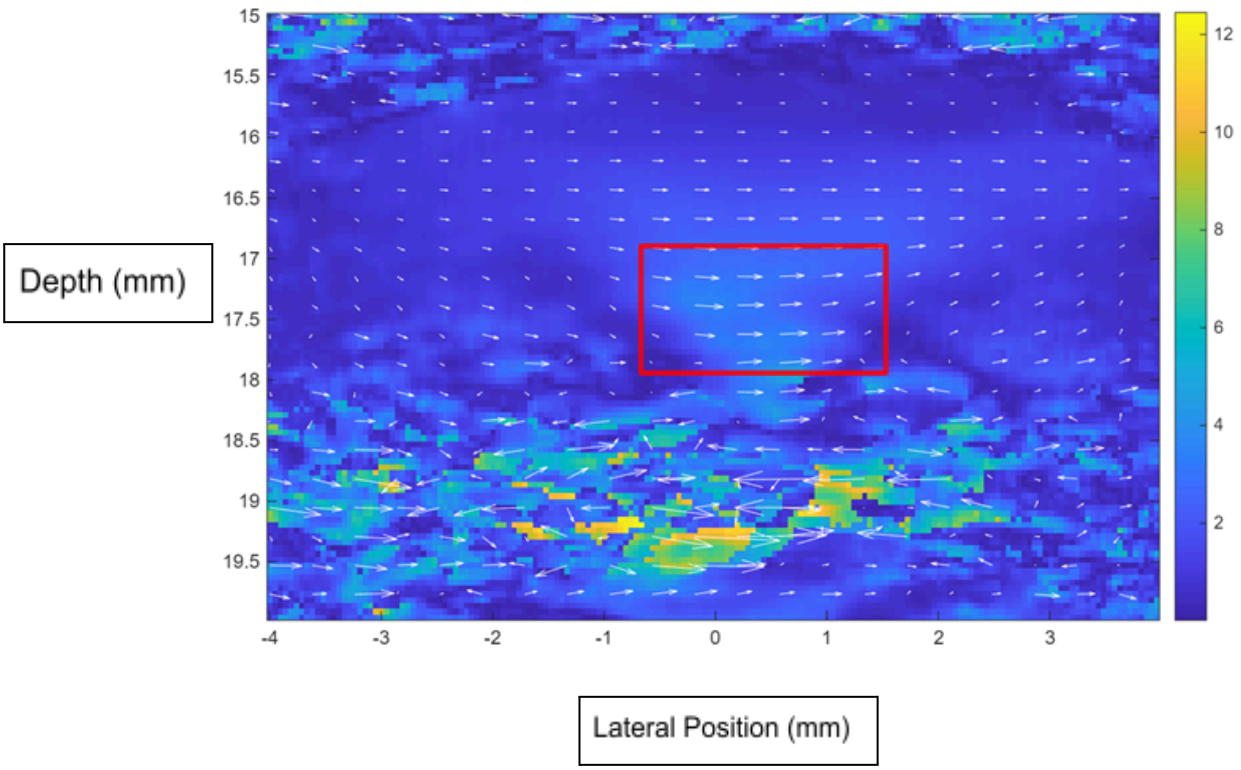

Figure G: Example image of a vector flow map detailing the magnitude and direction of flow in the phantom vein, with the chosen analysed region represented by the red box

With the arrows pointing in the direction of flow and the length of each representing the magnitude of the velocity of flow. The region chosen to find the mean and max velocities of flow and give an associated

standard deviation was the one shown by the 20 by 40 pixel red box at coordinates (70,36). These mean velocities were then compared at each wavelength for both fluids and to an expected mean flow velocity value calculated using equation x.

The maximum flow velocities for each fluid at each wavelength were then compared separately.

6. Results

These are the results obtained for all our key recordings during the experiment and data processing.

5.1 Doppler Shift, Laser Power and Intensity

The following graph shows the average doppler shift at each receive angle for all wavelengths for both fluids. This was also compared for expected shifts calculated using equation X:

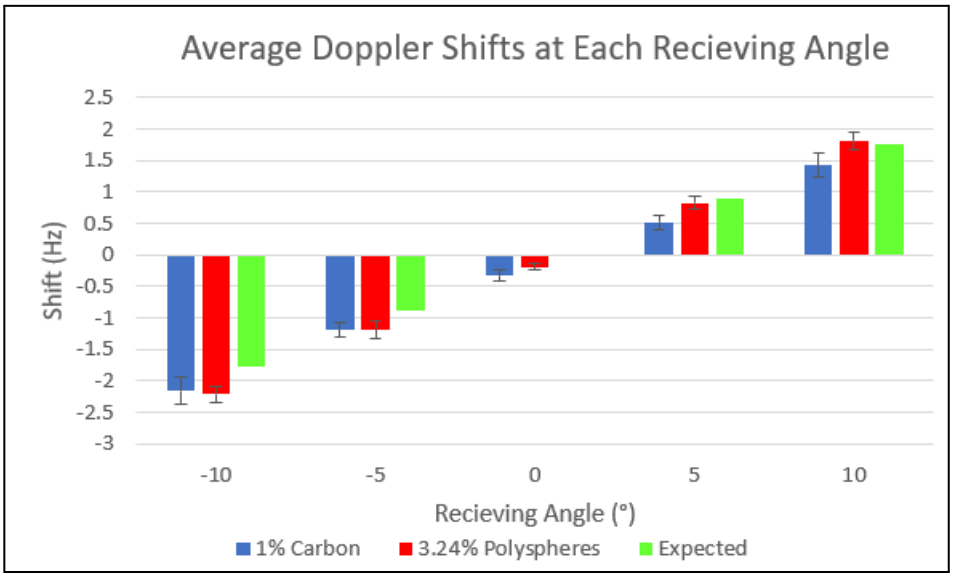

Figure I: Average doppler shift across our wavelength range of 680-800nm at each receiving angle compared to expected shifts calculated using equation x

From Figure I it's seen that 3.24% polyspheres give greater magnitude shifts compared to 1% carbon except at a receiving angle of 0°. Results for doppler shift of both fluids are shown to be larger in magnitude when the receiving angle is less than 0° compared to results above this value. Results below 0° are shown to be less precise compared to results above 0°.

Table T: Doppler shifts of 1% carbon and 3.24% polyspheres fluids averaged across the laser wavelength range of 680-880nm at each transducer receiving angle compared to the expected doppler shifts calculated using equation x

Receiving Angle (°)	Expected Shift (Hz)	1% Carbon Shift (Hz)	3.24% Polyspheres (Hz)
10	1.763	1.420 ± 0.1859	1.81 ± 0.1418
5	0.8848	0.520 ± 0.1180	0.827 ± 0.1056
0	6.22×10^{-16}	-0.3333 ± 0.08548	-0.187 ± 0.05888
-5	-0.8848	-1.187 ± 0.1198	-1.19 ± 0.1457
-10	-1.763	-2.157 ± 0.223	-2.22 ± 0.1256

Table T shows that shifts seen at receiving angles above 0° are shown to be more accurate compared to shifts below this value with the polyspheres fluid shown to give more accurate results across all angles.

The power of the laser prior to each recording and intensities found at each wavelength for both fluids is given below:

Table U: Maximum signal intensity and power of incident laser on the phantom at each wavelength as found from image reconstruction data. The intensity is normalised against laser power to account for fluctuations.

Wavelength (nm)	Laser Power (mW/cm^2)	Fluid	Signal Intensity	Normalised Signal Intensity
680	167	1% carbon	6112	36.60
		3.24% polyspheres	142.1	0.8509
720	163	1% carbon	4751	29.15
		3.24% polyspheres	304.2	1.866
760	156	1% carbon	2589	16.60
		3.24% polyspheres	132.4	0.8487
800	141	1% carbon	4509	31.98
		3.24% polyspheres	124.8	0.8851
840	117	1% carbon	6539	55.89
		3.24% polyspheres	87.63	0.7490
880	108	1% carbon	6552	60.67
		3.24% polyspheres	59.53	0.5512

Due to the varying power of the laser at each wavelength the intensity is normalised to see which is best at producing higher intensities relative to the laser power.

The maximum signal intensities found using the 1% carbon fluid are much higher compared to those found with the 3.24% polyspheres solution. These results are very inconsistent with each other. The highest ratio of maximum signal intensity to laser power found for the 1% carbon solution was 60.67 at a wavelength of 880nm, and for the 3.24% polyspheres solution this was 1.866 at 720nm.

5.2 Signal-to-Noise Ratio (SNR)

The following graph shows the SNR for both fluids at each wavelength:

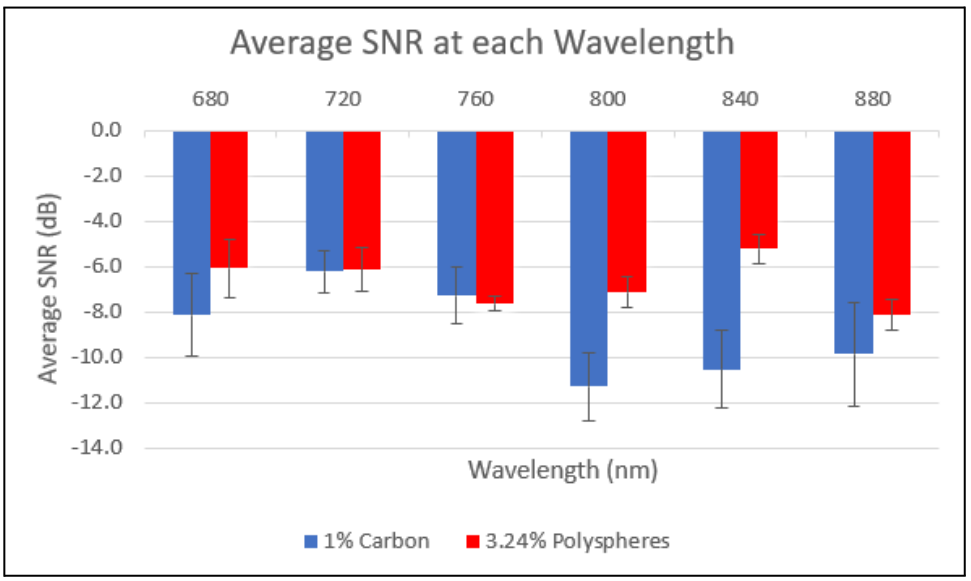

Figure J: Average SNR across all transducer receiving angles at each wavelength for 1% carbon and 3.24% polyspheres fluids plotted on a logarithmic scale

Figure J shows 3.24% polyspheres fluid has a less negative SNR compared to the 1% carbon fluid at all wavelengths, with an exception at 760nm. Results produced by the 3.24% polyspheres fluid are also shown to be more precise compared to results for the 1% carbon fluid. The best result for SNR for the 3.24% polyspheres fluid is -5.221 ± 0.6383dB at a laser wavelength of 840nm and for the 1% carbon fluid is -6.721 ± 0.9265dB

Table B: Average SNR over our range of wavelengths for 1% carbon and 3.24% polyspheres

Photoacoustic fluid	SNR (dB)

1% Carbon	-8.892 ± 2.364
3.24% Polyspheres	-6.721 ± 1.256

These results reinforce findings in Figure J, on average SNR is less negative and results are more precise for the 3.24% polyspheres fluid.

5.3 Vector Flow

This figure shows our results for mean flow velocity compared to the expected results from equation Z:

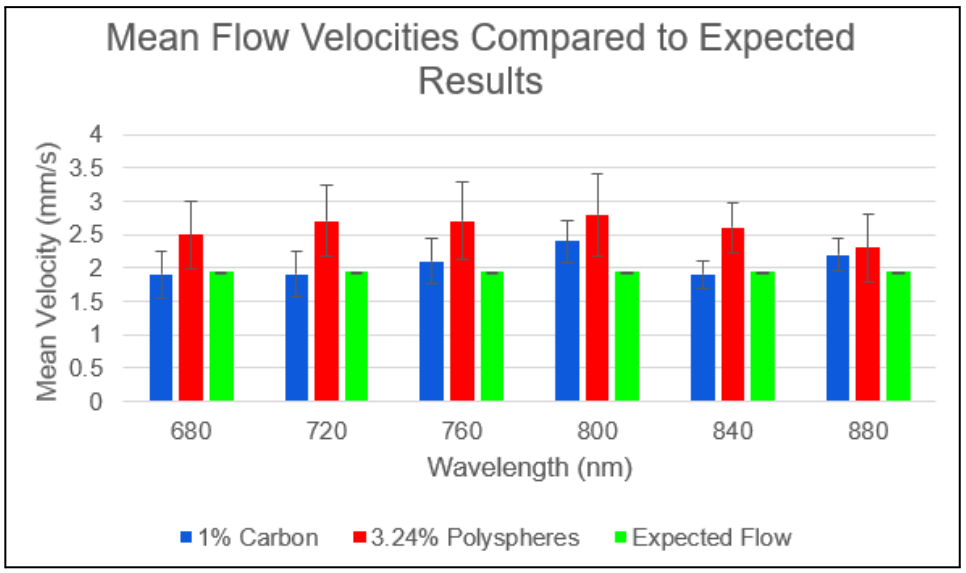

Figure K: Average velocity of flow seen by transducer at each wavelength ranging from 680-880nm for 1% carbon and 3.24% polyspheres fluid compared to expected results

Figure K shows that the average flow velocity of the 3.24% polyspheres fluid is higher compared to both the 1% carbon fluid and the expected results. They are also shown to be less precise compared to 1% carbon. The most accurate flow speed for 1% carbon fluid is 1.9 ± 0.1985mm/s at 840nm and for 3.24% polyspheres fluid is 2.3 ± 0.4997mm/s.

Table C: Average velocity of flow for 1% carbon and 3.24% fluids through the phantom vein compared to the expected result calculated using equation x

Photoacoustic fluid	**Flow Velocity (mm/s)**
Expected Flow	1.936 ± 0.004
1% Carbon	2.067 ± 0.2991
3.24% Polyspheres	2.600 ± 0.5238

Table C shows that both fluids over predict the expected flow, with the 1% carbon fluid shown to be more accurate compared to 3.24% polyspheres.

The Maximum flow velocities found at each wavelength are compared for both fluids in the following figure:

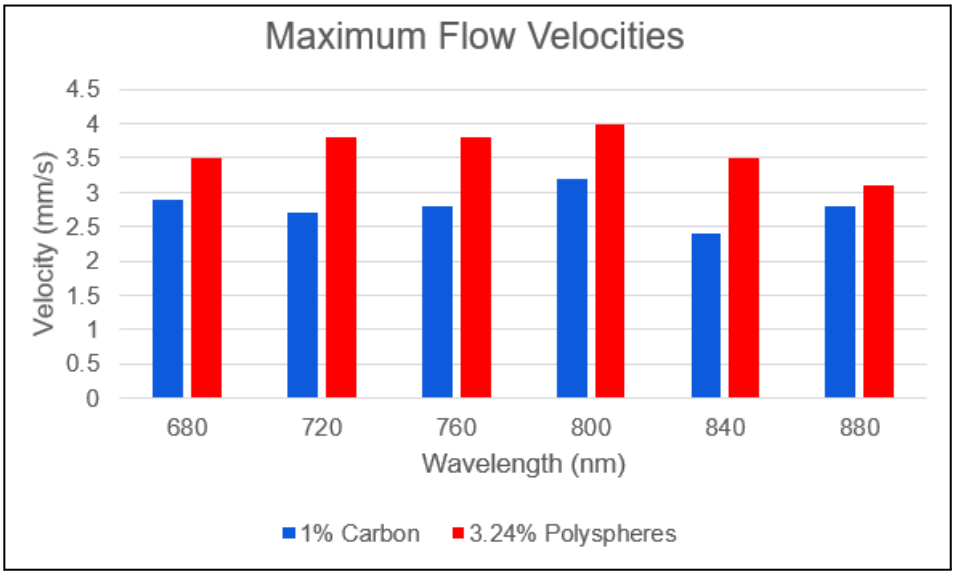

Figure L: Maximum flow velocities at each wavelength for 1% carbon and 3.24% polyspheres fluid

The maximum flow velocity for the 3.24% polyspheres fluid is shown to be larger than the 1% carbon fluid across all wavelengths.

7. Analysis of Results

7.1 Signal-to-Noise Ratio Performance

7.1.1 Polysphere Fluid SNR

7.1.2 Carbon Fluid SNR

7.1.3 SNR Comparison and Implications

The polysphere fluid demonstrated better SNR performance compared to the carbon fluid across all wavelengths we tested. The polysphere fluid achieved an average SNR of -6.7206 ± 1.2561dB as seen in **Fig.x** with its best performance at 840nm reaching -5.2210 ± 0.6383 dB as seen in **FIG.X.** The carbon fluid showed an average SNR of -8.8916 ± 2.3650 dB as seen in **FIG X,** with its best performance at 720nm having an SNR of -6.2374 ± 0.9265 dB as seen in **FIG X**.

This roughly 2dB difference in SNR is significant in PA. A higher (less negative) SNR means a stronger signal relative to background noise, which results in clever images and more reliable measurements. The polysphere fluids' better SNR performance suggests it would produce higher quality images and more easily distinguishable flow signals in practical PAI applications.

The carbon fluid also showed increased variability in its SNR measurements, with a standard deviation of 2.3650dB compared to 1.2561dB for the polyspheres. This indicates less consistent performance across different wavelengths. This contradicts our initial hypothesis that the carbon particles' stronger optical absorption would result in higher SNR values. The polysphere fluid's more uniform optical properties and reduced light scattering from its non-glassy surface likely contribute to its more stable and superior SNR performance.

7.2 Flow Velocity Accuracy

7.2.1 Polysphere Fluid Flow Measurements

7.2.2 Carbon Fluid Flow Measurements

7.2.3 Flow Accuracy Comparison and Implications

When comparing the measured flow velocities to our expected flow rate of 1.924 ± 0.004 mm/s, the carbon fluid demonstrated significantly better accuracy than the polysphere fluid. The carbon fluid achieved an overall average of 2.067 ± 0.29907 mm/s across all wavelengths, with its most accurate measurement at 840nm of 1.9 ± 0.19845 mm/s. The polysphere fluid showed an overall average of 2.6 ± 0.52388 mm/s across all wavelengths, with its best performance at 880nm being 2.3 ± 0.4997 mm/s.

The carbon fluids measurement at 840nm was within 1.25% of the expected flow rate, showing excellent accuracy for flow velocity measurements. In contrast, even the polyspheres' best measurement at 880nm was roughly 20% higher than the expected flow rate. Across all wavelengths, the carbon fluid consistently measured closer to the true flow velocity, with a total deviation of around 7.4% compared to the polyspheres 35% average deviation.

The polysphere fluid also exhibited considerably larger uncertainty in its measurements, with the standard deviations being nearly double that of the carbon fluid. This suggests that while the polysphere fluid produces clearer signals (as shown in the SNR results), these signals may be less accurate for quantitative velocity measurements. This presents an interesting trade-off between image quality and measurement accuracy. The reasons for the polysphere fluids systematic overestimation of flow velocity are unclear, but could potentially be related to its smaller particle size or its optical scattering properties affecting the doppler shift calculations.

7.3 Wavelength-Dependent Performance

7.3.1 Polysphere Fluid Wavelength Response

Polyspheres fluids have optical properties that better represent red blood cells in vivo due to absorption spectra of polystyrene being similar to that of oxygenated red blood cell haemoglobin and the red dye used to colour the spheres being similar to that of the pigment of red blood cells. We can look at the absorption coefficients of red blood cells and a 3µm polyspheres solution as seen in the supplementary material from Joanna (2016):

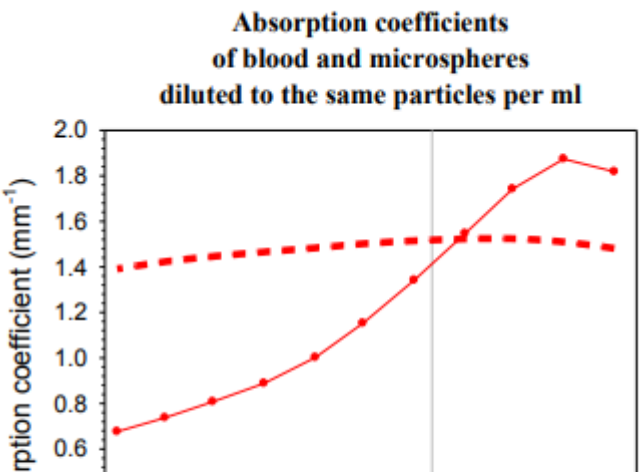

Figure T: Absorption coefficients for 3µm polyspheres solution compared to red blood cells with equivalent concentrations in de-ionized water over a wavelength range of 500-550nm

In Figure T we see that the absorption coefficients particularly agree between 530-540nm suggesting that in this particular range the absorption properties of the two fluids are very similar. This was not tested in our range of wavelength as we used a laser wavelength ranging from 680-880nm in 40nm steps.

7.3.2 Carbon Fluid Wavelength Response

7.3.3 Optimal Wavelength Selection

Both fluids showed distinct wavelength-dependent performance characteristics, with different optimal wavelength for SNR and flow accuracy measurements. For the polysphere fluid, 840nm provided the best SNR performance, while 880nm yielded the most accurate flow measurements. The carbon fluid showed its best SNR at 720nm and most accurate flow velocity at 840nm.

The difference in optimal wavelengths between the two fluids likely stems from their distinct optical properties. The polysphere fluid's red pigmentation is designed to mimic blood's absorption spectrum. Blood hemoglobin shows strong absorption peaks in the visible range at 540 and 576nm for oxygenated hemoglobin but also exhibits increased absorption in the near-infrared range above 700nm[1]. This aligns well

[1] https://pmc.ncbi.nlm.nih.gov/articles/PMC3005160

with the polysphere's best performance at 840-880nm. The fact we didn't observe performance at lower wavelengths is likely because our testing range started at 680nm, had we tested at lower wavelengths closer to blood's primary absorption peaks, we may have seen different optimal wavelengths. The carbon fluid's black pigmentation provides broader absorption across the spectrum but showed optimal performance at the shorter 720nm wavelength for SNR, suggesting different absorption characteristics compared to blood.

Interestingly, 840nm emerged as a particularly important wavelength in our experiments. It provided the best SNR for polyspheres and the most accurate flow measurements for carbon. This suggests that 840nm could be a good compromise wavelength when switching between fluids or prioritising different measurement goals. However, the fact that optimal wavelengths differ between SNR and flow accuracy measurements for the same fluid highlights the complexity of selecting experimental parameters in PAI. Depending on whether image quality or measurement accuracy is more critical for a given application, different wavelengths should be selected accordingly.

8.4 Overall Comparative Analysis

When evaluating both fluids across all metrics, it becomes clear that neither fluid emerges as a definitive winner for all photoacoustic imaging applications. Instead, each fluid presents distinct trade-offs in performance characteristics.

The polysphere fluid excels in signal quality, producing clearer images with better SNR performance across all wavelengths tested (average of -6.72dB compared to -8.89dB for carbon). This superior signal strength translates to better image clarity and more easily distinguished flow signals. However, this advantage in signal quality comes at the cost of measurement accuracy, with flow velocities consistently overestimated by ~35% compared to expected values.

The carbon fluid demonstrates the opposite performance profile. It achieves superior accuracy for flow velocity measurements, with velocity readings within 7.4% of expected values overall and near-perfect accuracy (within 1.25%) at 840nm. This makes it significantly more reliable for applications requiring precise velocity quantification. However, it produces weaker and less consistent signals, with both lower average SNR and greater variability across wavelengths compared to polyspheres.

The fundamental trade-off between these fluids is clear: signal quality vs measurement accuracy. The polysphere fluid provides better images but less accurate data, while the carbon fluid provides more accurate measurements but with weaker signals.

8. Error Analysis/Limitations

8.1 Systematic Errors

8.1.1 Wavelength-Dependent Laser Power Variations

A significant source of error in our photoacoustic measurements arose from wavelength-dependent variations in laser output power. Tunable laser systems tend to exhibit non-uniform power profiles across their operation wavelength range due to gain medium efficiency and properties [1,2]. These power fluctuations directly affect the amplitude of photoacoustic signals, potentially causing non-normality amongst our results. This was an important source of error throughout our experiment as we were using a variable wavelength laser and using multiple wavelengths. To ensure this was taken into account, we took power measurements for each of the wavelengths before doing the experiment, allowing us to correlate them with our results and ensuring no abnormalities arose from this specific issue.

8.1.2 Laser Alignment Uncertainty

The absence of an analytical method for laser alignment introduced potential limitations in experimental reproducibility. In photoacoustic imaging, precise spatial overlap between the optical excitation and acoustic detection regions is essential to ensure consistent signal amplitude and image quality. In this study, alignment was performed empirically by iteratively adjusting the laser position to maximise the measured signal amplitude and then fixing it in place at the observed maximum. While sufficient for single-session measurements, this approach reduces repeatability across multiple experimental sessions. Minor misalignments accumulated over time could significantly affect data reliability and comparability. Although this limitation did not impact our results, as all trials were conducted within a single day, the adoption of an analytical alignment procedure is recommended for experiments requiring repeated measurements.

8.2 Sample Stability and Preparation

8.2.1 Carbon Particle Settling

Carbon particle settling was both an immediate experimental concern and a long term stability issue that could affect measurement reproducibility and slow down experimentation. The carbon particles in our

carbon fluid have a higher density than the carrier fluid, causing them to settle over time due to gravity, in other words, they fall out of suspension. This causes a gradient of carbon and clear carrier fluid within the phantom and experimental setup. This introduces some issues with measurements, notably, if the carbon falls out of suspension, we are imaging the carrier fluid instead. To ensure that this was not a source of error within our experiment, we mixed the carbon fluid between each measurement and ensured it stayed in suspension during the measurement process. This is a relatively small issue, however it does raise some concerns with experimental viability, if you have to take time during the experiment to re-mix the fluid between each measurement, then using a blood mimicking fluid may be less convenient than thought. This is a major practical limitation in using the carbon fluid for blood mimicking and should be taken into account. In future experiments, a stabilising agent such as … could be used to prevent sedimentation and improve the practicality of the fluid.

8.2.2 Microsphere Staining

The microsphere fluid used was pink in colour to better mimic the optical properties of blood. After conducting trials with said fluid, we found it to be staining the outer edges of our phantom vein. This presents some major issues with running multiple experiments/trials and repeatability. If the outer edges of the phantom become too stained they can effectively “steal” the absorption from the flowing liquid and can result in measurement of the stagnant staining on the outer edges, making the results less reliable and accurate. This is because the dye molecules from the fluid bind to the phantom walls over time. Once bound, these molecules become stationary absorbers that contribute to the photoacoustic signal alongside the flowing fluid. Since we can't separate the signal from the stained walls versus the flowing fluid, our measurements of the fluid's properties become less accurate. This was not a significant issue for our limited number of trials, but could become problematic with extended use. For future repeatability, you may have to switch between phantoms or figure out a way of removing the staining from the inner vein between experiments.

8.2.3 Practical Limitations and Cost Considerations

Beyond performance metrics, the practical usability of each fluid presents significant limitations that impact their suitability for different experimental contexts. The polysphere fluid, while offering superior signal quality, comes at a substantial cost of $635 per 5mL, roughly 7 times more expensive than the carbon fluid. This cost difference can be prohibitive for laboratories conducting numerous trials or working with limited budgets, potentially restricting access to what is otherwise the better performing fluid in terms of image quality.

The carbon particle settling issue, as discussed in section 8.2.1, presents a major practical limitation during experimentation. The constant need to remix the fluid between measurements slows down the experimental process and introduces potential for inconsistency if mixing is insufficient. The handling

requirement makes the carbon fluid less convenient for extended measurement sessions or high volume experiments, despite its superior accuracy for flow measurements.

Similarly, the microspheres staining problem discussed in section 8.2.2 limits the long term repeatability of experiments using the polysphere fluid.As the phantom walls become stained over time, they act as stationary absorbers that cannot be distinguished from the flowing fluid signal, progressively degrading measurement accuracy. This necessitates either frequent phantom replacement (adding additional cost) or developing cleaning procedures, both of which add complexity to experimental workflows.

These practical limitations highlight that the choice between fluids cannot be made on performance data alone. The experimental context, including budget constraints, number of trials required, and timeframe for measurements, must be considered alongside the optical and flow measurement performance when selecting an appropriate blood mimicking fluid.

8.3 Signal Processing and Measurement Artifacts

8.3.1 Aliasing in Sampling

Aliasing was a potential issue within our experiment. When measuring vector flow it is important that the rate of flow is slow enough that you don't get any aliasing and noise artifacts being present in the measured signal. If the flow velocity is too high relative to our measurement rate, the velocity information can fold back and appear as flow in the wrong direction or at incorrect speeds. To avoid this, we ensured our flow rates were kept within a range where aliasing wouldn't occur by setting our flow rate to be half of the maximum measurable velocity (Vmax) for our given measurement system. If you wanted to measure faster flow this would need to be taken into account, although it is relative to the measurement system and laser used in experimentation, nonetheless an important consideration.

8.3.2 Environmental Noise

Photoacoustic measurements are sensitive to acoustic interference from environmental sources. This includes vibrations from building infrastructure, air conditioning systems, and electromagnetic interference. These noise sources add unwanted components to our signal that reduce the signal-to-noise ratio, particularly for weak signals from low absorber concentrations or at higher imaging depths. Without any acoustic shielding or active noise cancellation, these noise components will show up in results, potentially skewing the results or making them more difficult to read. As we did not have any reliable and realistic way to reduce or eliminate this noise, during our data processing, we ran the signal through a filter. This filter effectively removed the noise element from our results which allowed us to accurately measure and read results that were found as a direct result of our experiment. In future experiments, some active noise removal could be a good idea if you want good real time measurements.

8.4 Recommendations based on error sources.

Through this experiment, we have identified several key sources of error that affect the reliability and practicality of photoacoustic vector imaging when using blood mimicking fluids. These errors range from systematic issues like wavelength-dependent laser power variations and alignment uncertainty, to practical limitations like carbon particle settling and microsphere staining.

What became clear is that evaluating blood mimicking fluids isn't just about matching the best optical properties, considering the practicality of each fluid is just as important. A fluid that performs well optically but requires constant remixing or stains the phantom may be less suitable for extended measurements than one with slightly less ideal properties but easier workability.

In future experiments, it will be important to consider the discussed error sources and ideally come up with fluids for them. This could range from finding a way to clean the staining from the phantom, to finding an analytical method for laser alignment. All of the major sources of error found throughout this experiment are likely relatively straight forward to address and once fixed, blood mimicking fluids can become a more practical and reliable tool for vector flow imaging using photoacoustics.

Sources of error:

- Aliasing - potential issue
- Carbon falling out of suspension
- Spheres staining fluid
- Laser power changing with wavelength
- No analytical method for laser alignment
- Noise from other sources

Considerations:

- Carbon falling out of suspension
- Spheres staining fluid

Discussion Goes here

9. Conclusion